

Standard Operating Procedure

Camden CX-12 Plus Door Interface Relay – Integration with Kantech KT Controllers

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Applies To	Field Technicians, Installers
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Owner	Apex – Operations
Approved By	_____

NOTE ON THIS DOCUMENT:

The CX-12 Plus specifications, DIP switch modes, and terminal numbers below are drawn directly from Camden Door Controls' official CX-12 Plus Installation Instructions (Rev. March 2020). Kantech terminology (Lock outputs, Relay outputs, REX inputs) is drawn from Kantech's published KT-400 installation documentation. No manufacturer publishes a combined Camden-to-Kantech wiring diagram — the integration steps below apply verified facts from both manufacturers' own documentation, but the technician **MUST** confirm the exact output type (open-drain Lock output vs. Form C Relay output) and terminal location on the specific Kantech controller model and firmware at the site before wiring, using that controller's own installation guide.

1. Purpose

This procedure defines the standard method for integrating a Camden CX-12 Plus Door Interface Relay with a Kantech KT-series access control controller (KT-1, KT-2, KT-4, or KT-400), typically to prevent an automatic door operator from attempting to open a door before the electric lock or strike has released.

2. Scope

Applies to Apex technicians installing or servicing a CX-12 Plus relay used as an interface between a Kantech KT-series controller's lock/relay output and an automatic door operator, electric strike, or maglock.

3. Device Overview

Camden CX-12 Plus — Door Interface Relay

Provides a make/break relay (switching network) function that eliminates automatic door operators binding against a still-locked strike or maglock, by delaying the door operator until the lock has de-energized (or energized, depending on mode).

- Input voltage: 12/24V AC/DC, non-polarity sensitive (Terminals 1 & 2)
- Inputs: 2 Wet (powered, 3–30V AC/DC, optically isolated) and 2 Dry (N.O.), all electrically isolated from each other
- Outputs: 2 x Form C (SPDT) relays, 3A @ 30VDC rated, 100,000 operations electrical life
- 8 selectable operating modes via 3 DIP switches (SW1–SW3)
- 3 adjustable time delays (potentiometers), 1–30 seconds each

Kantech KT-Series Controller (general)

KT-400 and similar KT-series controllers provide dedicated, supervised Lock outputs (open-drain, one per door) for driving an electric strike or maglock directly, separate from general-purpose Relay outputs (Form C, dry contact) used for auxiliary functions such as this integration. Each door also has a REX (Request to Exit) input and a Door contact input. Confirm which output type (Lock vs. Relay) and which physical terminal is appropriate for this integration using the specific controller model's installation guide before wiring — terminal layouts differ between KT-1, KT-2, KT-4, and KT-400.

4. Recommended Integration Approach

The most common application for this pairing is: the Kantech controller controls the electric lock/strike as usual, and the CX-12 Plus — configured in Access Control Mode (Mode 2) — delays the automatic door operator until the lock has released, then fires the operator. This prevents the operator motor from straining against a still-locked door.

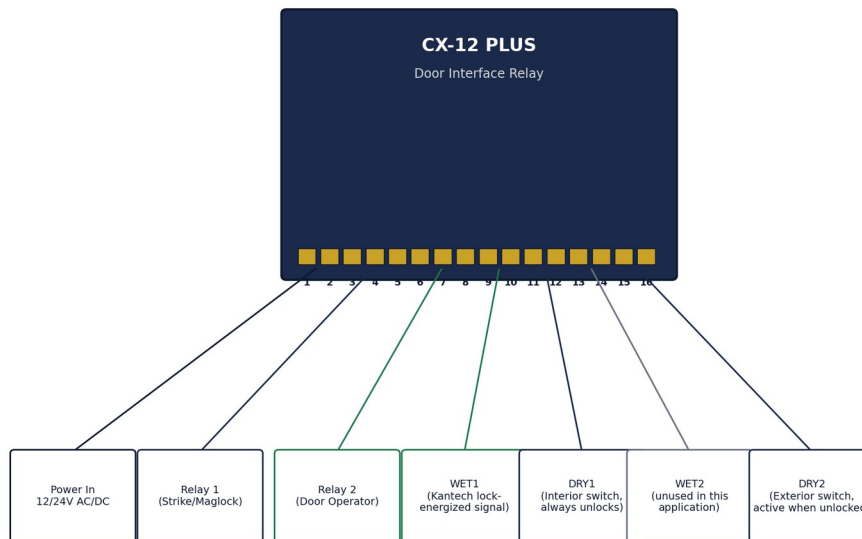
- If the Kantech output being used provides a powered, maintained signal while the lock is unlocked (e.g., tapped from the lock power circuit while energized): wire it to CX-12 Plus WET1 (Terminals 9 & 10).
- If only a dry (Form C) Relay output is available on the Kantech controller: do not wire it directly to WET1, since WET1 requires an actual powered signal, not a dry contact. Either tap a powered signal elsewhere in the lock circuit, or reconsider which CX-12 Plus mode fits the dry-contact signal available (see Camden's Mode 1 — Momentary — for a dry-contact-driven application instead).
- The automatic door operator connects to CX-12 Plus Relay 2 (N.O. & COM, Terminals 6 & 7).
- An interior wall switch (always unlocks/opens, regardless of Kantech status) may be wired to CX-12 Plus DRY1 (Terminals 11 & 12), if the application calls for one.
- An exterior activating switch (only active while the lock is energized) may be wired to CX-12 Plus DRY2 (Terminals 15 & 16), if the application calls for one.

5. CX-12 Plus Terminal Reference

- Terminals 1 & 2: Power input, 12/24V AC/DC, non-polarity sensitive
- Terminals 3, 4, 5: Relay 1 — N.O., COM, N.C. (strike typically uses N.O./COM 3&4; maglock typically uses COM/N.C. 4&5)
- Terminals 6, 7, 8: Relay 2 — N.O., COM, N.C. (door operator connection)
- Terminals 9 & 10: Wet 1 input (powered, isolated)
- Terminals 11 & 12: Dry 1 input
- Terminals 13 & 14: Wet 2 input (powered, isolated)
- Terminals 15 & 16: Dry 2 input

CX-12 Plus — Terminal Reference (Access Control Mode / Mode 2)

Terminal numbers and function per Camden Door Controls CX-12 Plus Installation Instructions



Confirm exact Kantech output type (open-drain Lock vs. Form C Relay) and terminal on the specific controller model before wiring — see SOP-LV-012, Section 3-4.

6. Setting Access Control Mode (Mode 2)

Set DIP switches SW1=ON, SW2=OFF, SW3=ON for Access Control Mode.

- Do not apply power to the CX-12 Plus until all wiring is complete and the DIP switches are set.
- WET1 (Terminals 9 & 10): connect the Kantech powered/maintained lock-energized signal.

- DRY2 (Terminals 15 & 16): exterior activating switch, if used — active only while the lock is energized.
- DRY1 (Terminals 11 & 12): interior switch, if used — always unlocks and opens the door regardless of WET1 status.
- Relay 2 (Terminals 6 & 7, N.O./COM): wire to the automatic door operator's activation input.

7. Timing Adjustment

Adjust the three onboard potentiometers to set the delay between the lock releasing and the operator firing:

- Most modern door operators have their own built-in hold-open timers. Where this is the case, set the CX-12 Plus to send only a brief momentary pulse (1–2 seconds) to the operator, and let the operator's own timer handle hold-open duration.
- Walk-test the door after adjustment. The operator should not begin moving before the lock has visibly released.

8. Verification

- Confirm the lock releases fully before the door operator begins moving — no binding or strain on the operator motor.
- Test both the Kantech-driven unlock path and any interior/exterior switches wired to the CX-12 Plus.
- Confirm the door operator holds open long enough for normal pedestrian egress/ingress.
- Confirm the Kantech controller still reports door/lock status correctly in EntraPass after the integration — the CX-12 Plus should not interfere with the controller's own door position or REX monitoring.
- Record the CX-12 Plus mode, DIP switch settings, and timer values on the site configuration sheet.

9. Notes

Do not wire life-safety or fire/smoke-release devices through the CX-12 Plus — per Camden's own installation instructions, safety devices must be wired directly to the door operator's control box, not through this relay.

If the integration does not behave as expected after following this procedure, do not continue adjusting settings speculatively. Re-confirm the Kantech output type and terminal against that controller model's installation guide, and re-confirm the CX-12 Plus DIP switch and terminal wiring against Camden's manual before escalating.

References

- Camden CX-12 Plus Door Interface Relay — Installation Instructions, Rev. March 26, 2020 (Part No. 40-82B241)
- Kantech KT-400 Ethernet Four-Door Controller Installation Manual
- SOP-LV-010 — KT-1/KT-2/KT-4/KT-400 Controller Default & Configuration
- SOP-LV-011 — KTES Door Entry Default & Configuration